

Machine Learning-Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

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Abstract

Real estate firms serve very different kinds of buyers, yet many still market to them as one group. This study applies unsupervised machine learning to a client and transaction dataset supplied for the Parcl Co. Limited project to find out whether distinct buyer segments exist and how they differ. Ten thousand property listings (7,305 sold) were linked to 2,000 clients, and purchase behaviour was aggregated into one record per buyer. After cleaning, encoding and scaling, 13 features were clustered with K-Means and validated against Ward hierarchical clustering. The silhouette score favoured $K = 3$ (0.212), but $K = 4$ (0.145) was adopted to match the project brief and yield more actionable groups. The four segments are: Value Buyers (662 clients), Satisfied Mid-Market buyers (638), Premium Buyers (649) and a small group of Bulk Multi-Unit Buyers (51). They differ mainly in satisfaction, spend, unit price, floor area and number of units purchased. They do not differ meaningfully in age, loan use, investment intent or client type. Cluster separation is modest, and the paper reports this openly. A Streamlit dashboard was built to let analysts explore the segments interactively.

Keywords: buyer segmentation, K-Means, hierarchical clustering, silhouette score, real estate analytics, customer profiling, Streamlit

1. Introduction

1.1 Background

Property companies attract home buyers, investors, companies and overseas purchasers, and these groups differ in budget, motivation, financing and expectations. Market segmentation, the practice of dividing a market into groups that respond differently to marketing effort, has been recognised as a core marketing strategy since Smith (1956). When a company treats all buyers alike, marketing spend is spread thinly, property recommendations are generic and high-value clients are hard to identify.

Clustering algorithms offer a data-driven alternative to hand-drawn segments. Instead of assuming that buyers fall into predefined categories, clustering groups clients by how similar their recorded characteristics and behaviour are, and the analyst then interprets each group.

1.2 Problem Statement

Parcl Co. Limited does not currently have a data-driven view of (a) the types of property buyers it serves, (b) the motives behind purchases across demographic groups, (c) geographic

differences in buyer behaviour and (d) financing patterns. The consequences are inefficient marketing spend, generic recommendations and poor targeting of investors.

1.3 Objectives

1. Clean and combine the client and property datasets into one analysis-ready table with one row per buyer.
2. Engineer purchase-behaviour features (units bought, total spend, average price, average area).
3. Segment buyers with K-Means and validate the result with hierarchical clustering.
4. Choose the number of clusters with the elbow method and the silhouette score.
5. Profile each segment by purchase intent, geography, financing and demographics.
6. Deliver the results as an interactive Streamlit dashboard and translate them into recommendations.

1.4 Contribution and Structure

Besides producing segments, this paper documents what the data does not support. Segment types suggested in the project brief, such as corporate buyers and first-time buyers, did not emerge from this dataset, and the evidence for that is presented in Section 6. Section 2 reviews the methods used, Section 3 describes the data, Section 4 explores it, Section 5 explains the methodology, Section 6 reports results, Section 7 describes the dashboard, and Sections 8 to 11 cover recommendations, limitations, future work and conclusions.

2. Methods Background

Two standard clustering methods were used. K-Means groups observations around centroids that are recomputed until they stabilise; it is fast and easy to interpret but sensitive to feature scaling. Ward hierarchical clustering merges the closest clusters step by step, which reveals nested structure and gives an independent check on K-Means. The number of clusters was judged with the elbow method and the silhouette coefficient (Rousseeuw, 1987), and the two solutions were compared with the adjusted Rand index (Hubert and Arabie, 1985). All models were built with scikit-learn (Pedregosa et al., 2011).

3. Data

3.1 Sources

Two tables were supplied. The **properties** table has 10,000 rows and 9 columns (listing, tower, transaction date, unit category, unit number, floor area, sale price, listing status and client reference). The **clients** table has 2,000 rows and 12 columns (client ID, client type, name, date of birth, gender, country, region, acquisition purpose, satisfaction score, loan applied and referral channel). Names were not used in the analysis.

3.2 Cleaning and Validation

- **Duplicates:** no duplicated rows were found in the properties table, each listing ID was unique, and no client ID was duplicated in the clients table. Repeated client references in the properties table were retained because one client can legitimately buy several units.
- **Missing values:** the only column with missing values was the client reference, with 2,695 nulls (26.95% of rows). A cross-tabulation showed that every unit with status Available had no client and every Sold unit had one (7,305 sold, 2,695 available). The nulls therefore represent unsold inventory, not data errors. They were not imputed, because inventing buyers would corrupt the segmentation. Available units were excluded from buyer analysis.
- **Value checks:** sale prices ranged from 97,403 to 736,652 (mean 344,375) and floor areas from 411 to 1,957 sq ft (mean 1,140). No zero or negative values were found.
- **Labels:** categorical labels were standardised to lower case with consistent spelling, and the loan indicator was coded 0/1.
- **Linkage:** all 7,305 sold units matched a client record, giving 2,000 unique buyers.

3.3 One Row per Client

Clustering needs one record per buyer, so sold transactions were aggregated by client to create units_bought, total_spend, avg_price, avg_area and towers_used. Age was computed from the date of birth. Each client bought between 3 and 13 units (mean 3.65). Most bought 3 (932 clients) or 4 units (948 clients), while 120 bought five or more. This long right tail is what allows a distinct bulk-purchase group to appear.

4. Exploratory Data Analysis

4.1 Client Profile

Attribute	Summary
Client type	94.8% individual, 5.2% company
Gender	50.6% male, 49.4% female
Age	25 to 95 years, mean 55.9
Acquisition purpose	69.2% personal home, 30.8% investment
Loan applied	36.8%
Referral channel	55.2% website, 35.2% agency, 9.6% existing client
Satisfaction score	1 to 5, mean 3.03 (SD 1.41)
Country	USA 76.9%, then UK, Canada, Germany, France, Belgium, Mexico, Australia, Russia, Denmark
Regions	57 distinct regions

The client base is dominated by individual buyers in the United States, and companies are a small minority. These facts shape what any clustering can reveal. The distributions are shown in Figure 2.

4.2 Relationships between Variables

The correlation heatmap (Figure 1) shows that the purchase variables move together: units bought and towers used correlate at 0.93, units bought and total spend at 0.71. Age, satisfaction score and loan status are close to uncorrelated with every other variable. Spending, in other words, is not explained by who the client is in this dataset. Average unit price and average floor area are also very closely related (0.96). Because towers_used duplicates units_bought, it was left out of the model.

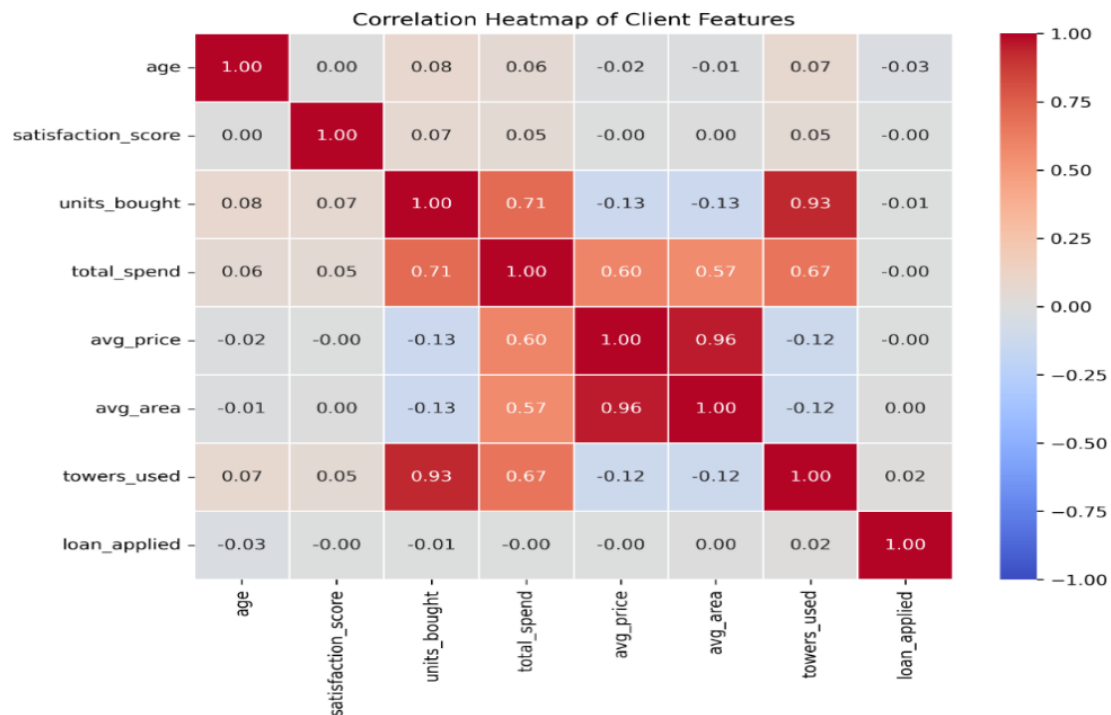


Figure 1. Correlation heatmap of client features.

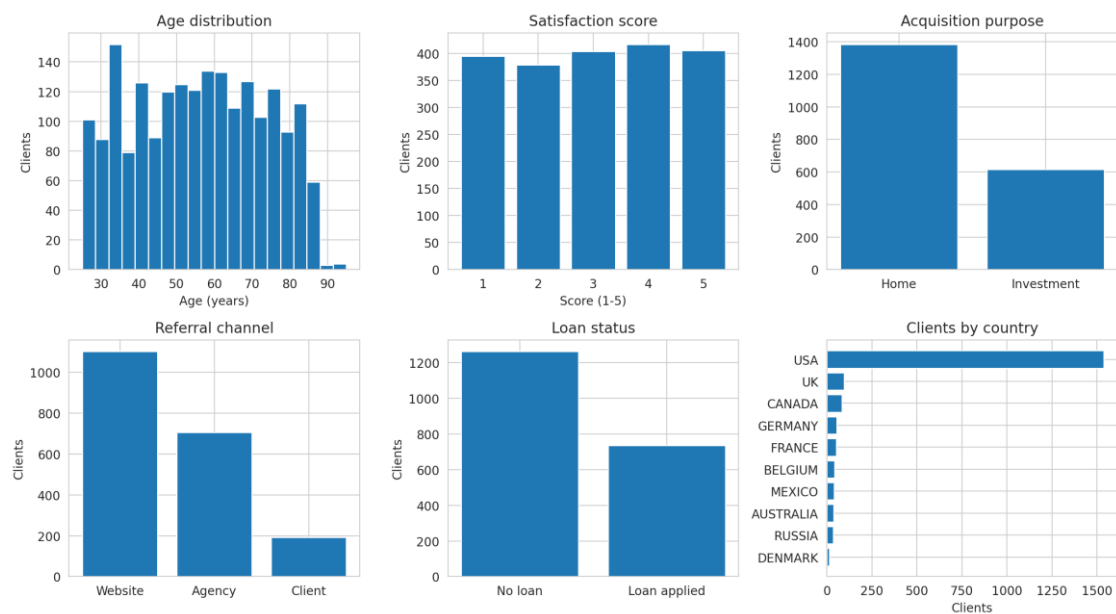


Figure 2. Distributions of age, satisfaction, purpose, loan status and referral channel.

5. Methodology

5.1 Feature Engineering

Because money amounts are right-skewed, total_spend and avg_price were log-transformed with $\log(1 + x)$. The numeric features were age, satisfaction score, units bought, log total spend, log average price and average floor area.

5.2 Encoding

One-hot encoding was applied to the low-cardinality nominal variables client type, acquisition purpose, referral channel and gender, with a single column kept for binary variables. Loan applied was already 0/1. Label encoding was also applied to the categorical fields to produce compact numeric identifiers for reporting, but those columns were not used in the models because integer codes imply a false ordering that distance-based clustering would treat as meaningful.

Country (10 values) and region (57 values) were deliberately kept out of the clustering input. One-hot encoding them would add 67 binary columns next to only six numeric features, so geography would dominate the distances and the result would reflect location, not buyer behaviour. Geography is used instead to interpret the clusters afterwards, which matches the profiling step in the project brief.

5.3 Scaling

Numeric features were standardised with StandardScaler so that no variable dominated through its units alone. The final matrix had 2,000 rows and 13 columns with no missing values.

5.4 Clustering and Model Selection

K-Means (k-means++ initialisation, 10 restarts, random seed 42) was run for $K = 2$ to 9. Inertia was recorded for the elbow plot and the silhouette score for each K . Ward hierarchical clustering was then applied to the same matrix, and the two solutions were compared with a cross-tabulation and the adjusted Rand index.

5.5 Cluster Interpretation

Each cluster was profiled on size, age, satisfaction, units bought, spend, price, floor area, loan rate, investment share, client type, referral channel and country. Cluster means were also expressed as a percentage difference from the average client, which makes the defining traits visible at a glance.

6. Results

6.1 Choosing K

K	Inertia	Silhouette
2	11,249.8	0.201
3	10,034.8	0.212
4	9,151.4	0.145
5	8,339.0	0.143
6	7,830.2	0.136
7	7,434.1	0.133
8	7,077.6	0.132
9	6,754.5	0.132

The elbow curve declines steadily and has no sharp bend (Figure 3). The silhouette is highest at $K = 3$ and drops after that (Figure 4). Scores in this range indicate weak cluster separation, which is typical of behavioural data without well-separated groups.

$K = 4$ was nevertheless selected. The project brief calls for four buyer segments, and four groups give finer, more actionable profiles than three. The cost is a lower silhouette (0.145 against 0.212), and this trade-off should be kept in mind when reading the segments.

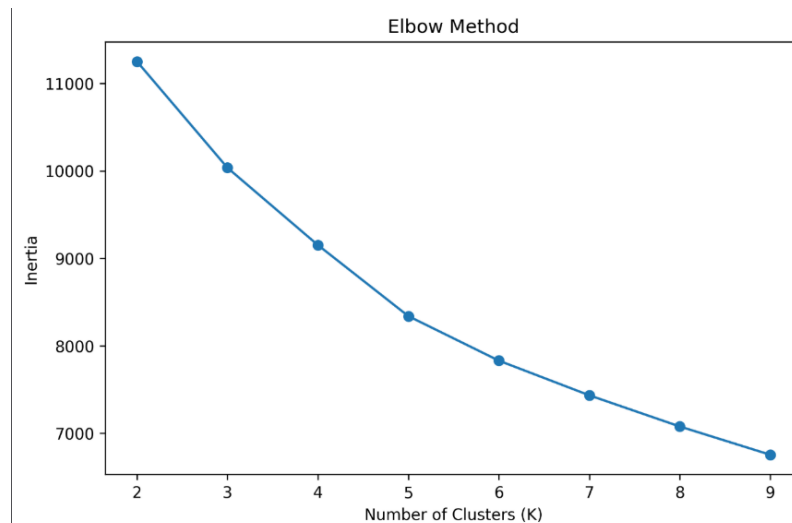


Figure 3. Elbow plot.



Figure 4. Silhouette score by K.

6.2 The Four Segments

Segment	Clients	Age	Satisfaction	Units	Avg total spend	Avg unit price	Avg area (sq ft)	Loan rate
Value Buyers	662	56.1	2.41	3.42	965,581	283,514	946	37%
Satisfied Mid-Market	638	55.4	4.15	3.75	1,264,156	338,880	1,122	37%
Bulk Multi-Unit Buyers	51	64.6	3.49	7.29	2,397,660	330,963	1,095	31%
Premium Buyers	649	55.6	2.53	3.51	1,467,988	421,277	1,382	36%

Figure 5 shows how each segment differs from the average client.

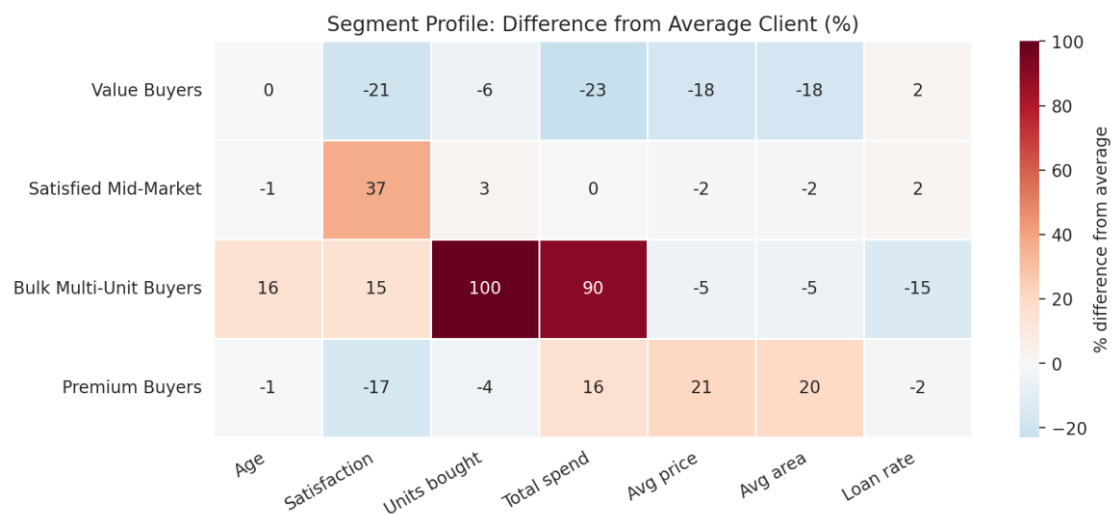


Figure 5. Segment profile heatmap, percentage difference from the average client.

Value Buyers (662). The lowest-spending group: the cheapest and smallest units, the lowest total spend, and the lowest satisfaction of any segment.

Satisfied Mid-Market (638). Mid-range units and spend, but by far the highest satisfaction (4.15 out of 5). This group looks like the most content clients.

Premium Buyers (649). The most expensive and largest units (average price 421,277, average 1,382 sq ft). Despite the highest prices, satisfaction is below average (2.53), which is a retention risk among the clients with the most valuable homes.

Bulk Multi-Unit Buyers (51). A small group that buys about twice as many units as others (7.3 against about 3.5) and has the highest total spend. Its members are older (64.6 years), less likely to use loans (31%), and 71% were referred through the website, against 55% elsewhere. Only 4% were referred by existing clients.

6.3 Validation with Hierarchical Clustering

Ward clustering with four clusters scored a silhouette of 0.116, lower than K-Means (0.145). The adjusted Rand index between the two partitions was 0.27, which indicates only moderate agreement. The cross-tabulation shows one clear point of agreement: 38 of the 51 Bulk Multi-Unit Buyers fall in a small hierarchical cluster of 38, so both methods single out this group. Most Value Buyers (535 of 662) also map to a single hierarchical cluster. The remaining two K-Means segments are split across several hierarchical clusters, which suggests that the boundary between them is soft.

6.4 What Does Not Separate the Segments

Several variables show almost no difference across segments:

Variable	Range across the four segments
Investment purpose	28% to 35%
Loan applied	31% to 37%
Company clients	3.9% to 5.9%
Average age (excluding bulk buyers)	55.4 to 56.1 years

This matters for the project brief. Segments described as Corporate Buyers (companies purchasing multiple units) and First-Time Buyers (younger and loan-dependent) are not present here: no segment is company-heavy, and none is younger or more loan-dependent than the rest. The segments are driven by spend, unit characteristics and satisfaction, not by demographics or financing. Geography adds little as well. The USA accounts for 77% of clients, and shares in small countries rest on very few clients (for example, Denmark has only 15), so country-level patterns should be treated as indicative at best.

7. Interactive Dashboard

A Streamlit application was built to make the segmentation explorable by non-technical users. It has four modules, matching the project requirements:

- **Segmentation Overview:** cluster distribution and a spend-versus-satisfaction view.
- **Investor Behaviour:** purpose, loan use, spend, price and referral channel by segment.

- **Geographic Analysis:** a world map, segment mix by country and a country-region breakdown.
- **Segment Insights:** descriptive statistics, a difference-from-average heatmap and a drill-down for each segment.

Users can filter by country, region, acquisition purpose, client type, segment and age range, and download the filtered data. The dashboard and its code are available at:

- **Live dashboard:** <https://ramchandan2059-hue-financial-analytics-real-estate-app-hnjszo.streamlit.app/>
- **GitHub repository:** <https://github.com/ramchandan2059-hue/Financial-Analytics-Real-Estate>

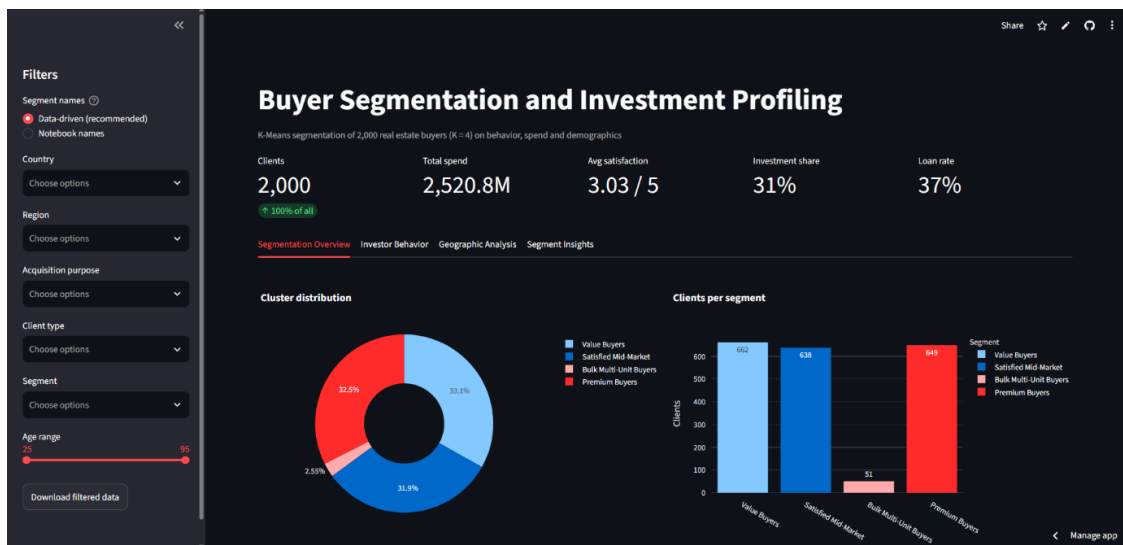


Figure 6. Dashboard screenshot.

8. Business Recommendations

Segment	Suggested action
Premium Buyers	Prioritise retention. Assign relationship managers, survey the cause of low satisfaction, and offer early access to larger or high-end units. This group represents the highest unit prices.
Satisfied Mid-Market	Use as advocates. Introduce referral incentives, since only about 10% of clients arrive through existing clients, and collect testimonials.
Value Buyers	Investigate dissatisfaction first. Examine pricing, unit quality and service for smaller, lower-priced units before spending on acquisition. Consider financing and upgrade paths.
Bulk Multi-Unit Buyers	Treat as key accounts. Offer portfolio-level service, bundled purchases and dedicated contacts. Because the group is small and website-sourced, target similar profiles through digital channels.

More generally, the low satisfaction among two of the four segments, including the one with the priciest units, is the clearest improvement opportunity in the data. Marketing based on age or loan status would not separate these buyers.

9. Limitations

1. **Weak cluster structure.** Silhouette scores of 0.145 ($K = 4$) and 0.212 ($K = 3$) show overlapping segments, and the agreement between K-Means and hierarchical clustering is moderate (ARI 0.27). The segments are a useful description of tendencies, not sharply bounded groups.
2. **Choice of K.** $K = 4$ was selected to follow the project brief and not because it scored best.
3. **Limited features.** Only a few numeric variables are available, and several (age, satisfaction, loan status) are nearly uncorrelated with behaviour. No income, wealth or property-type information was provided.
4. **Undocumented data provenance.** No documentation explains how satisfaction scores were collected. The near-zero correlations between demographics and spending are consistent with a simulated or heavily anonymised dataset, so the findings are best read as a demonstration of method and a source of hypotheses, not as proof about real buyers.
5. **Geographic imbalance.** With 77% of clients in the USA, comparisons across countries and regions are unreliable for small groups.
6. **Small bulk segment.** The 51-client group is clearly distinct but too small for firm statistical conclusions.
7. **Static snapshot.** Segments describe one point in time and were not tested for stability over time.

10. Future Work

- Test mixed-type methods such as K-Prototypes or Gower distance with hierarchical clustering, which handle categorical variables more directly.
- Add features such as income, transaction timing, purchase frequency and property type.
- Evaluate cluster stability with bootstrapping and alternative algorithms (Gaussian mixtures, DBSCAN).
- Build a classifier that assigns new clients to segments automatically.
- Link segments to outcomes such as repeat purchase and referral behaviour to test their business value.

11. Conclusion

This study combined 10,000 property listings with 2,000 client records, built one behavioural profile per buyer, and segmented the buyers with K-Means, validated by hierarchical clustering. Four segments were identified: Value Buyers, Satisfied Mid-Market buyers, Premium Buyers and a small group of Bulk Multi-Unit Buyers. They are separated by spend, unit price, floor area, units purchased and satisfaction, and not by age, loans, investment intent or client type. Cluster separation is modest, and several segment types suggested in the project brief were not found. The Streamlit dashboard turns these results into a tool that supports targeted retention, referral and key-account strategies, and the limitations above indicate where richer data would sharpen the picture.

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